SEQUENCES LADDER 4 OF 9

Newton's formula: the table as a rule

The first entry of each difference row is a coefficient, and binomial coefficients turn the table into a formula.

1 / THE RULE AND THE PICTURE

Read down the left-hand diagonal and multiply by binomial coefficients

u_n	5	12	23	38	57
Δ	7	11	15	19	
Δ^2	4	4	4		

Take the first entry of each row, the red diagonal $5, 7, 4$, and use it with the binomial coefficients $\binom{n-1}{k}$:

$$u_n = u_1 + \Delta u_1 \binom{n-1}{1} + \Delta^2 u_1 \binom{n-1}{2} + \cdots$$

Here $u_n = 5 + 7(n-1) + 4\binom{n-1}{2} = 2n^2 + n + 2$.

2 / THE BINOMIAL BUILDING BLOCKS

Each $\binom{n-1}{k}$ is a column of Pascal's triangle

$n =$	1	2	3	4	5	6
$\binom{n-1}{0}$	1	1	1	1	1	1
$\binom{n-1}{1}$	0	1	2	3	4	5
$\binom{n-1}{2}$	0	0	1	3	6	10
$\binom{n-1}{3}$	0	0	0	1	4	10

The block $\binom{n-1}{0}$ is always 1, $\binom{n-1}{1} = n-1$, $\binom{n-1}{2} = \frac{(n-1)(n-2)}{2}$ is $0, 0, 1, 3, 6, 10$, and each block is zero until its own turn, which is why the formula fits the first terms one at a time.

3 / TWO WORKED EXAMPLES

Quadratic and cubic

For $5, 12, 23, 38, 57$ the top diagonal is $5, 7, 4$, so $u_n = 5 + 7(n-1) + 2(n-1)(n-2) = 2n^2 + n + 2$, which agrees with the peeling method.
For $4, 15, 48, 115, 228$ the diagonal is $4, 11, 22, 12$, so $u_n = 4 + 11(n-1) + 22\binom{n-1}{2} + 12\binom{n-1}{3} = 2n^3 - n^2 + 3$, found without ever subtracting a power.

4 / WHY IT WORKS

Differencing a binomial lowers its index

Pascal's rule $\binom{m+1}{k} - \binom{m}{k} = \binom{m}{k-1}$ says that taking the difference of $\binom{m}{k}$ gives $\binom{m}{k-1}$, so $\Delta\binom{n-1}{k} = \binom{n-1}{k-1}$, exactly as differentiating $x^k/k!$ gives $x^{k-1}/(k-1)!$.
Applying Δ to the whole formula reproduces the row of differences, so it must be the right rule.

5 / RUNNING THE TABLE BACKWARDS

The formula also gives u_0

Putting $n = 0$ in the first example uses $\binom{-1}{k} = (-1)^k$ and gives $5 - 7 + 4 = 2$, the term that comes before 5, since running the first differences $7, 11, \ldots$ backwards gives $7 - 4 = 3$ and $5 - 3 = 2$.
The same rule with a formula in n lets us ask for $u_{1.5}$ or u_{-3}, which a list of terms cannot.

6 / TRY THESE

Use the diagonal

(a) For $1, 4, 9, 16, 25$ the diagonal is $1, 3, 2$, so $u_n = 1 + 3(n-1) + 2\binom{n-1}{2} = n^2$. **(b)** For $3, 5, 9, 15, 23$ the diagonal is $3, 2, 2$, so $3 + 2(n-1) + 2\binom{n-1}{2} = n^2 - n + 3$.
(c) What is the top diagonal of $2, 2, 2, 2$? $2, 0$, so the rule is constant.

ASIDE / THE CALCULUS TWIN

Newton's formula in binomials is the discrete twin of the Taylor series in powers $x^k/k!$, with differences in place of derivatives.

LINKS TO OTHER SHEETS

The polynomial interpolation sheet derives the same Newton form from divided differences, and the Worpitsky page shows the jet at $n = 0$.

Convention. $\binom{n-1}{k} = \frac{(n-1)(n-2)\cdots(n-k)}{k!}$; the top diagonal is $u_1, \Delta u_1, \Delta^2 u_1, \ldots$; Newton's forward difference formula.